

MiroFish on Free Tier — Full Run Summary

Project: Atlas AR GTM prediction

Date: June 2026

Simulation ID: `sim_4fa4b86a352a` · Project ID: `proj_74a5fa4cd43e`

Goal

Run a live MiroFish prediction for Atlas AR: seed → graph → simulation → report, using free or low-cost LLM APIs.

Steps Completed

Step	Status	Notes
Clone MiroFish to <code>d:\AI\agency-agents\MiroFish</code>	Done	From GitHub
Install deps (Node + Python)	Done	<code>pip</code> failed on Windows; fixed with <code>uv sync</code> (~132 packages)
Configure <code>.env</code> (Zep + LLM keys)	Done	Zep key worked throughout
Seed upload + ontology generation	Done	After fixes (see obstacles)
Graph build	Done	~7 minutes
Agent preparation	Done	~3 minutes
40-round simulation	Done	Completed on backend
Report generation	Not finished	Blocked by LLM quotas

Models Used

Simulation (steps 1–5) — COMPLETED

Model	Provider	Result
<code>gemini-2.0-flash</code>	Google Gemini	Blocked — free tier quota = 0 for this model
<code>gemini-2.5-flash</code>	Google Gemini	Used to complete full simulation

Model	Provider	Result
Second Gemini key	Google Gemini	Works for single calls; blocked for report (20 req/day limit)

Report only (step 6) — ALL BLOCKED

Model	Provider	Result
<code>gemini-2.5-flash</code>	Gemini	20 requests/day free limit
<code>gemini-2.5-flash-lite</code>	Gemini	Same daily limit
<code>openai/gpt-4o-mini</code>	OpenRouter	402 — no credits purchased
<code>meta-llama/llama-3.2-3b-instruct:free</code>	OpenRouter	Upstream rate limits (429)
<code>openrouter/free</code>	OpenRouter	Best progress (~14 min into report), then daily + per-minute limits
<code>nvidia/nemotron-3-nano-30b-a3b:free</code>	OpenRouter	JSON works; used as fallback
<code>nvidia/nemotron-nano-9b-v2:free</code>	OpenRouter	Often returns empty content
<code>nvidia/llama-nemotron-rerank-v1-1b-v2:free</code>	OpenRouter	Wrong model type — reranker only, not chat

Obstacles (Chronological)

1. Setup and environment

- **Missing LLM key** — Zep alone is not enough; MiroFish needs an OpenAI-compatible LLM for every step.
- `pip install` failed on Windows — `camel-oasis` not available via pip; fixed with `uv sync`.
- **No Docker** — native Windows path required `uv`.
- **Backend would not start from script** — health check used wrong URL; fixed to `http://127.0.0.1:5001/health`.
- **Duplicate processes on port 5001** — old backends kept stale model config until killed.

2. Gemini (simulation + early report attempts)

- `gemini-2.0-flash`: free tier limit = 0 → switched to `gemini-2.5-flash`.
- **Ontology JSON truncated** at 4096 tokens → increased to **16384** in `ontology_generator.py`.
- **Simulation timeout in client script** — simulation actually **finished on backend**; client poll limit was too short (~30 min).
- **Gemini free tier: 20 LLM requests/day per model** — too low for report agent (~20+ calls with tool use).

3. OpenRouter (report attempts)

- Paid models (`gpt-4o-mini`): 402 Insufficient credits — account never purchased credits.
- Free models: 429 rate limits — per-minute (16/min) and per-day caps.
- OpenRouter message: Add 10 credits to unlock 1000 free model requests/day.
- Empty responses from some free models (`None` content) → crashed report agent until model fallback chain was added in `llm_client.py`.

4. Wrong model choice

- NVIDIA Llama Nemotron Rerank VL 1B V2 is a reranking model (`/rerank` API), not a chat model — cannot generate reports.

What Worked vs What Did Not

Worked (free tier)

- Zep graph memory
- Gemini 2.5 Flash → full 40-round simulation
- OpenRouter free models → short calls, partial report progress
- Manual prediction doc (`PREDICTION-REPORT.md`) when engine could not finish

Blocked (free tier)

- Gemini → 20 req/day (simulation used most of budget)
- OpenRouter paid → no credits
- OpenRouter free → daily + per-minute caps mid-report
- Nemotron Rerank → wrong API (rerank, not chat)

Artifacts Created

File	Purpose
<code>docs/atlas-ar/mirofish/SEED-ATLAS-AR.md</code>	Simulation seed
<code>docs/atlas-ar/mirofish/PREDICTION-REPORT.md</code>	Manual/spec-driven prediction
<code>docs/atlas-ar/mirofish/runs/CHECKPOINT-live-run.json</code>	Resume point for report-only
<code>scripts/run-mirofish-prediction.mjs</code>	Full pipeline orchestrator
<code>scripts/resume-mirofish-report.mjs</code>	Report-only resume script

No live `REPORT.md` was saved under `runs/` — the report step never completed.

Bottom Line

Free tier can run a MiroFish simulation if you use the right Gemini model (`gemini-2.5-flash`) and accept setup friction on Windows.

Free tier cannot reliably finish the report — the report agent is a multi-step LLM workflow (tool calls + reflection) that needs dozens of API calls. Gemini's 20/day limit and OpenRouter's free caps are both too tight.

Cheapest path to finish: add ~\$10 OpenRouter credits or use a direct OpenAI key, then run report-only:

```
node d:\AI\agency-agents\atlas-webxr\scripts\resume-mirofish-report.mjs
```

No re-simulation needed — `sim_4fa4b86a352a` is already done on the backend.